DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (A)
Osmania University, Hyderabad-07

CLASS TEST-II

Subject : Operating Systems

Class: B.E (CSE-AIML) WP — **Max Marks: 30**
Semester: V — **Duration:1 ½ hr**
Academic Year: 2026 — **Date :04-05-2026**

S. No.	PART A Answer All Questions (Q5 x 2M = 10 M)	Marks	BT	CO
1.	What are the four conditions that a deadlock can arise?	3	3	1
2.	What is logic bomb? How it may manifest itself?	2	3	3
3.	Define protection & security.	2	2	3
4.	Mention the best fit, worst fit and first fit allocation strategies	3	1	2
	PART B Answer Any Two Questions (02 x 10M = 20M)			
5.	Illustrate the use of the banker's algorithm, consider a system with five processes Po through P4 and three resource types A, B, and C. Resource type A has ten instances, resource type B has five instances, and resource type C has seven instances. (see table below) Check whether the system is in safe state or not . Justify a) Suppose now that process P1 requests one additional instance of resource type A and two instances of resource type C. Whether this request be immediately granted or not. b) a request for (3,3,0) by P4 can be granted or not. Justify	10	3	1
6.	a) Consider the below reference string, illustrate the Optimal and LRU page replacement algorithms. 7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1	5	3	2
	b) Describe the process of encryption. What is public key encryption?	5	2	3
7.	a) Explain the implementation of access matrix.	5	3	3
	b) Elaborate paging technique.	5	1	2

Table for Question 5:

	Allocation	*Max*	*Available*
	A B C	*A B C*	*A B C*
P_0	0 1 0	7 5 3	3 3 2
P_1	2 0 0	3 2 2	
P_2	3 0 2	9 0 2	
P_3	2 1 1	2 2 2	
P_4	0 0 2	4 3 3	